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Final Year Project Report

Hard Joshi
2512658

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Supervisor: Arish Siddiqui
CN6000

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Abstract

This is where your abstract goes. Remember, it must be 500 words or less. It is NOT an introduction. It needs three elements: Purpose, Methodology, and Outcome. [cite: 2, 3]

Acknowledgments

Here you can thank the people, including your supervisor, who have helped you with your project. [cite: 5]

Contents

Abstract i
Acknowledgments ii
1 Introduction 1
2 Literature Review 2
2.1 The Central Argument: A Gap Three Decades Wide 2
2.2 Stream I - The M&A Paradox and the Failure of Valuation Theory 3
2.2.1 The Empirical Record: Stationarity of Failure 3
2.2.2 Behavioural Distortions: Hubris as Label Noise 3
2.2.3 Market Timing and the Stock-Driven Wave 4
2.2.4 The Epistemological Barrier: Structural Blindness 4
2.3 Stream II - The Tabular Paradigm: An Asymptotic Ceiling 4
2.3.1 The Econometric Era: Linearity Bias and Its Persistence 4
2.3.2 The Machine Learning Escalation: Why XGBoost Hits a Wall 5
2.3.3 The Deep Learning False Dawn: MLPs and the Independence Assumption . 6
2.4 Stream III - The Semantic Turn: Powerful Tools, Wrong Targets 6
2.4.1 The Bag-of-Words Era and the Loughran-McDonald Benchmark 6
2.4.2 The Transformer Revolution and the Limits of Standard FinBERT 7
2.4.3 Why Standard FinBERT Cannot Recover the Full Textual Signal 8
2.5 Stream IV - The Topological Turn: Network Structure as Irreducible Signal 9
2.5.1 The Economic Foundation: Supply Chain Momentum 9
2.5.2 Corporate Network Centrality and Value 9
2.5.3 Why XGBoost Cannot Recover the Topological Signal (Even With Graph Features) 10
2.5.4 Heterogeneous Graphs: Why Edge-Type Semantics Matter 11
2.6 The Measurement Foundation: Event Study Methodology 11
2.6.1 The Market Model and Its Implementation 11
2.6.2 Critical Assumptions and Their Limitations 12
2.6.3 Justification for Binary Classification 12
2.7 The Multimodal Imperative: Why Fusion Is the Only Exit 13
2.7.1 Complementary Variance and the Failure of Mono-Modal Models 13
2.7.2 Why HeteroGraphSAGE Fusion Is the Only Logical Exit 13
2.7.3 Architectural Choice: Late Fusion and Its Justification 14
2.8 Synthesis: Gap Table and Research Hypotheses 14
2.8.1 Formal Research Hypotheses 16
3 Methodology 17
3.1 Introduction 17
3.2 Research Philosophy and Design Logic 17
3.2.1 Philosophical Positioning 17
3.2.2 Quantitative Empirical Strategy 18
3.3 Data, Sampling, and Target Construction 18
3.3.1 Data Sources 18
3.3.2 Sample Construction 18
3.3.3 Event Study Label Design 19

3.3.4 Defence of the Event Window ... 19
3.3.5 Defence of the Market Model: Why Single-Factor OLS Is Sufficient at This Scale ... 20
3.4 Preprocessing and Leakage Control ... 20
3.4.1 Prevention of Forward-Looking Data Leakage ... 20
3.5 Modality-Specific Feature Construction ... 21
3.5.1 Financial Features ... 21
3.5.2 Textual Features and the PCA Critique ... 21
3.5.3 Graph Construction and HeteroGraphSAGE ... 22
3.6 Model Architecture and the Late-Fusion Defence ... 23
3.6.1 Why Joint End-to-End Training Was Rejected ... 23
3.6.2 The Overfitting Paradox and Inductive Transfer Framing ... 23
3.6.3 Why SHAP Becomes Methodologically Central ... 24
3.7 Evaluation Design and Hypothesis Testing ... 24
3.7.1 Model Ladder and Ablation Logic ... 24
3.7.2 Hypothesis Tests ... 24
3.8 Limitations Faced and Defended ... 25
3.8.1 Event-Window Contamination Risk ... 25
3.8.2 Simplified Expected Return Model ... 25
3.8.3 Information Loss from PCA Compression ... 25
3.8.4 Transfer Misalignment in Frozen Embeddings ... 25
3.9 Ethics, Reproducibility, and Finality of Implementation ... 26
4 Findings and Discussion ... 27
4.1 Introduction ... 27
4.2 Model Ablation Results ... 27
4.2.1 The Ablation Wall ... 27
4.2.2 Hyperparameter Tuning: An Honest Admission ... 28
4.2.3 Linear Regression: Confirming the Binary Framing ... 28
4.3 The M2 Reversal: Naive NLP Destroys Predictive Value ... 28
4.4 Hypothesis Test Results ... 29
4.4.1 H1: Topological Alpha (Supported) ... 29
4.4.2 H2: Semantic Divergence (Supported) ... 30
4.4.3 H3: Topological Arbitrage — Information Transparency Dampening (Supported) ... 31
4.5 SHAP Decomposition: Economic Credibility ... 31
4.5.1 The Global SHAP Manifold ... 31
4.6 Translating AUC into Financial Impact ... 32
4.7 Limitations of the Findings ... 33
4.7.1 Performance Ceiling ... 33
4.7.2 The Unresolved Text Architecture Problem ... 33
4.7.3 H3 Visualisation vs. Formal Causal Claim ... 33
4.8 Chapter Summary ... 33
Bibliography ... 35
A Code ... 41

B Evaluation Include an evaluation of both product and process. Be objective. Reflect on what worked and what did not. 42
C Conclusion Reflect on key findings, limitations, and future opportunities. 43
D Initial Project Proposal 44
E Final Project Proposal 45
F Application for Approval of Research Activities 46
G Client Consent Form 47

Chapter 1: Introduction

Chapter 2: Literature Review

2.1 The Central Argument: A Gap Three Decades Wide

Existing quantitative frameworks for predicting post-acquisition synergy are fundamentally constrained by a shared architectural assumption: they treat the firm as an isolated data point, devoid of context, history, or structural position. Financial ratio models, machine-learning classifiers, and even transformer-based NLP pipelines all process firms as nodes without edges - objects detached from the industrial ecosystems that define their actual value. This literature review builds a sustained critical argument across four interconnected knowledge streams:

1. **Stream I - The M&A Paradox and the Failure of Valuation Theory**, which diagnoses *why* synergy prediction is difficult and demonstrates that the root cause is structural information loss, not computational insufficiency.
2. **Stream II - The Tabular Paradigm**, covering the full history of quantitative M&A models from logistic regression through gradient boosting and early deep learning, demonstrating that each generation inherits the same fatal independence assumption - and establishing the precise ceiling beyond which single-modality tabular models cannot advance.
3. **Stream III - The Semantic Turn**, tracing the evolution of financial NLP from bag-of-words lexicons to transformer architectures, critically demonstrating that despite increasing sophistication, the field systematically misdirects powerful tools at deal *occurrence* rather than deal *outcome*, and that even FinBERT implementations fall into a "Tabular Trap" the moment their embeddings are flattened into scalar vectors.
4. **Stream IV - The Topological Turn**, engaging with the supply chain finance, corporate networks, and graph neural network literatures to establish that network topology carries *irreducible* economic signal about synergy potential - signal that is mathematically unrecoverable by any non-graph model regardless of its parameter count.

The review concludes that these four streams converge on a single logical exit strategy: a heterogeneous graph architecture that simultaneously fuses financial fundamentals (Block A), domain-specific textual embeddings (Block B), and supply-chain/competitor graph topology (Block C) within a unified predictive framework - a configuration that, as of 2025, no published study has deployed for post-merger synergy *outcome* prediction. Every section below is therefore not a neutral survey but a *motivated critique*: prior work is evaluated for what it structurally *cannot* achieve, and each limitation is mapped directly to one of the three hypotheses tested in this dissertation.

2.2 Stream I - The M&A Paradox and the Failure of Valuation Theory

2.2.1 The Empirical Record: Stationarity of Failure

Mergers and acquisitions remain the primary mechanism for global capital reallocation, yet their empirical track record constitutes one of the most durable anomalies in corporate finance. Martynova and Renneboog (Martynova and Renneboog, 2008a) documented that between 70% and 90% of acquisitions fail to generate value for acquirer shareholders across a full century of global takeover activity - a finding replicated across multiple independent meta-analyses (Clayton M. Christensen, Richard Alton, Curtis Rising and Andrew Waldeck, 2011). Critically, this failure rate has exhibited *statistical stationarity* across distinct economic regimes - from the conglomerate wave of the 1960s through the dot-com era and post-financial-crisis consolidations - implying that the root cause is structural rather than cyclical (Martynova and Renneboog, 2008a).

Bradley, Desai, and Kim (Bradley, Desai and Kim, 1988) established that while target shareholders captured average announcement-date abnormal returns of approximately 30%, acquirer shareholders experienced systematic losses, yielding combined synergy gains averaging only 7.4% of combined pre-deal firm value. This gap between the value created for targets and the value transferred (or destroyed) for acquirers is the precise economic surplus that a reliable predictive model should identify *ex ante*. The persistence of this destruction in the face of progressively more sophisticated computational due diligence tools motivates the epistemological question this study addresses: what information, structurally absent from conventional models, would close this predictive gap?

2.2.2 Behavioural Distortions: Hubris as Label Noise

The first explanatory layer is behavioural. Roll's Hubris Hypothesis (Roll, 1986a) argues that acquiring managers systematically overestimate their ability to extract value, treating the target's efficient market price as an underestimate rather than an unbiased signal. This creates the "Synergy Trap" identified by Sirower (Mark L. Sirower, 1997), wherein the premium paid mathematically requires performance improvements that are unattainable in the majority of cases.

From a machine learning perspective, hubris introduces systematic *label noise* into training datasets: if acquisition premiums partly reflect executive overconfidence rather than genuine synergy potential (Roll, 1986a), models trained on transaction values are fitting a signal contaminated with behavioural error. This contamination is *irreducible* - no amount of regularisation or hyperparameter tuning can remove bias that is encoded into the target variable itself. This motivates the use of Cumulative Abnormal Return (CAR) as the prediction target rather than deal premia or accounting synergies, since CAR represents the market's *ex post* revision of expectations and is less susceptible to managerial narrative.

2.2.3 Market Timing and the Stock-Driven Wave

A complementary source of label contamination is identified by Shleifer and Vishny (Andrei Shleifer and Robert W. Vishny, 2003), whose stock-market-driven acquisitions model demonstrates that overvalued acquirers rationally exploit inflated equity as acquisition currency during market peaks. Under this framework, a substantial proportion of M&A waves are motivated by relative misvaluation rather than operational complementarity - the observed CAR reflects market correction dynamics rather than synergy realisation. Grossman and Stiglitz (Grossman and Stiglitz, 1980) established the theoretical bound here: in a world of costly information acquisition, markets cannot be fully informationally efficient, and persistent information asymmetries around deal announcements are a natural equilibrium. This "bounded efficiency" framing justifies the premise that pre-announcement features can contain predictive signal about CAR direction - information costs prevent instantaneous arbitrage of the signal this study attempts to extract.

2.2.4 The Epistemological Barrier: Structural Blindness

Even setting aside behavioural distortions, the verification mechanism is structurally deficient. Angwin (Angwin, 2001) documented that conventional due diligence operates in organisational silos - financial, legal, and operational teams independently assess their respective domains without modelling interaction effects. This orthogonal approach misses multiplicative risks: strong cash flows combined with a fragile single-source supplier dependency produce a combined exposure that sums to near-zero value; linear due diligence passes both in isolation and misses the conjunction.

Akerlof's (George A. Akerlof, 1970) "Market for Lemons" framework provides the theoretical grounding for this structural blindness: mandated disclosures (10-K filings) are insufficient to eliminate information asymmetry because targets can strategically obscure ecosystem vulnerabilities within the noise of standard reporting formats (Hansen, 1987). The implication is precise: reliable synergy estimation requires information sources that cannot be strategically manipulated - specifically, the verifiable topology of the firm's external network of suppliers, customers, and competitors. This forms the primary theoretical motivation for Block C in this study's architecture.

Stream I gap: The M&A failure rate is predictable from structural information that is systematically absent from existing valuation models. The following three streams diagnose exactly which information each prior paradigm discards - and why.

2.3 Stream II - The Tabular Paradigm: An Asymptotic Ceiling

2.3.1 The Econometric Era: Linearity Bias and Its Persistence

The first generation of quantitative M&A predictors - Palepu (Palepu, 1986a), Barnes (Barnes, 1990a) - employed logistic regression and multiple discriminant analysis on financial ratios. These models operate under a "Linearity Bias": the assumption that financial metrics bear monotonic relationships to synergy potential. Barnes (Barnes, 1990a) demonstrated empiri-

cally that this assumption fails for leverage: moderate debt signals fiscal discipline; high debt signals distress. This creates a non-monotonic, context-dependent relationship that linear discriminants fundamentally cannot represent.

More critically, Palepu's (Palepu, 1986a) own out-of-sample validation found that acquisition targets could not be reliably predicted beyond chance - a result that exposed the information deficit in ratio-based models rather than any implementation flaw. The pseudo-R^2 ceiling below 0.10 documented consistently across econometric studies (Betton, Eckbo and Thorburn, 2008a) is not a calibration failure; it is mathematical evidence that financial ratios do not contain sufficient information to reconstruct synergy outcomes. This ceiling is the first benchmark any successor model must demonstrably surpass.

2.3.2 The Machine Learning Escalation: Why XGBoost Hits a Wall

The emergence of ensemble methods promised to resolve the non-linearity problem. Zhang et al. (Zhang *et al.*, 2024) deployed random forests and gradient-boosted decision trees on financial ratio vectors, reporting accuracy gains over logistic baselines. These improvements are real - but they are *bounded* and *misleading*. Four structural problems afflict these approaches, each of which XGBoost cannot overcome regardless of depth or regularisation configuration:

Problem 1 - The Independence Ceiling. Both random forests and XGBoost operate on rows of a feature matrix where each row corresponds to one deal in isolation. The model learns per-deal statistical patterns but has no mechanism to propagate information across related deals - if two acquirers share a supplier and that supplier is distressed, both deals share latent risk that tabular models cannot detect. This is not a data engineering problem; it is an architectural impossibility.

Problem 2 - Survivorship Bias. M&A deal samples drawn from completed transactions systematically undersample value-destructive deals that were abandoned or never announced, biasing estimated coefficients toward the characteristics of deals that *proceeded* rather than deals that *succeeded* (Betton, Eckbo and Thorburn, 2008a). The consequence is that reported out-of-sample accuracy reflects the characteristics of deal survival, not value creation - a fundamental misdirection that no ensemble architecture can correct without restructuring the sampling frame.

Problem 3 - Look-Ahead Contamination. Feature engineering pipelines for M&A ML studies frequently construct ratios from annual filings that post-date the deal announcement, embedding future information into the training set. Zhang et al. (Zhang *et al.*, 2024), for instance, discretise financial variables into categorical bins without explicit documentation of fiscal year alignment with announcement dates. This look-ahead contamination inflates validation accuracy beyond any economically realisable level.

Problem 4 - Feature Engineering as Crystallised Bias. Discretising continuous financial variables into categorical bins before tree construction (a common pipeline step in (Zhang *et al.*, 2024)) encodes human analyst heuristics into irreversible feature representations. The

model learns from a discretised proxy for the raw signal - an automated version of the same heuristic reasoning employed by the human analysts whose systematic failures motivated building the model in the first place.

The combined effect of these four problems is an asymptotic accuracy ceiling: once the linear, rank-order, and non-parametric signal in tabular financial features has been captured, no further improvement is achievable through more sophisticated tabular models. This ceiling is *not* a function of computational power - it is a function of the information content of the feature space.

2.3.3 The Deep Learning False Dawn: MLPs and the Independence Assumption

The introduction of deep MLPs and LSTMs appeared to dissolve the linearity constraint. Elhoseny, Metawa et al. (Elhoseny *et al.*, 2022) achieved 95.8% accuracy on financial distress prediction using an Ameliorative Whale Optimization Algorithm-tuned deep neural network - a result that superficially suggests deep learning has resolved the prediction problem. It has not. The model processes each firm as a standalone entity, consuming individual financial metrics without any mechanism for inter-firm information exchange. Impressive accuracy on financial distress classification does not transfer to M&A synergy prediction: the former asks "is this firm failing?" from its own time series; the latter asks "will combining two firms create more value than they possess separately?" - a fundamentally relational question that non-relational architectures cannot answer.

The same architectural constraint applies to even the most sophisticated MLP implementations. A model with 10 layers and 10,000 parameters, trained on a vector of 50 financial ratios, still operates in a 50-dimensional space that is topologically flat: it contains no representation of whether the acquirer's top customer is about to go bankrupt, whether the target and acquirer share a fragile common supplier, or whether the deal creates a bottleneck in the industrial network. These omissions are not correctable by adding more layers.

Stream II gap: Every generation of tabular model - logistic regression, random forest, XGBoost, MLP, LSTM - inherits the independence assumption from its predecessor. The asymptotic ceiling is not a function of model complexity but of feature space topology. Breaking through it requires a graph operator (Stream IV). Before reaching that conclusion, Stream III examines why NLP approaches, despite initially appearing to transcend the tabular paradigm, ultimately reproduce the same structural failure.

2.4 Stream III - The Semantic Turn: Powerful Tools, Wrong Targets

2.4.1 The Bag-of-Words Era and the Loughran-McDonald Benchmark

The seminal contribution to financial NLP is Loughran and McDonald (Loughran and McDonald, 2011a), who demonstrated that general-purpose sentiment lexicons systematically misclassify financial language: terms like "liability," "risk," and "obligation" carry negative sentiment in general English but are neutral legal descriptors in financial filings. Their domain-specific word lists corrected this systematic mislabelling and established a critical

methodological principle: financial text requires finance-specific analytical tools. This insight remains foundational and is directly inherited by this study's use of FinBERT over general BERT.

However, the bag-of-words (BoW) paradigm itself is architecturally insufficient beyond this correction. BoW models are "compositionally blind" (Loughran and McDonald, 2011a): they cannot distinguish "We have eliminated our exposure to risk" from "We have significant exposure to risk," because both sentences produce near-identical word frequency vectors while conveying directly opposite strategic signals. In the high-stakes context of M&A filings, where a single conditional clause ("subject to regulatory approval") can invert the meaning of a multi-billion dollar liability disclosure, this ambiguity is not marginal but catastrophic. Despite this, recent work continues to extend the BoW approach: Demers, Wang and Wu (Demers, Wang and Wu, 2024) construct frequency-based lexicons for human capital disclosures; Acheampong et al. (Acheampong *et al.*, 2025) proxy financial constraints using lexicon-based indices; Garcia, Hu and Rohrer (Garcia, Hu and Rohrer, 2020) develop finance dictionaries through market-based validation - all approaches that encode compositional blindness into increasingly refined vocabulary lists.

Hoberg and Phillips (Hoberg and Phillips, 2016) represent the most structurally sophisticated BoW application: their text-based network industries classification (TNIC) used product-description cosine similarity from 10-K filings to construct time-varying firm similarity networks, demonstrating that textual proximity predicts competitive dynamics and deal *likelihood* with meaningful accuracy. This study directly inherits the semantic proximity methodology from Hoberg and Phillips (Hoberg and Phillips, 2016) - but redirects it from deal occurrence to deal outcome, and replaces TF-IDF vectors with contextual FinBERT embeddings.

2.4.2 The Transformer Revolution and the Limits of Standard FinBERT

Devlin et al.'s (Devlin *et al.*, 2018a) BERT architecture rendered static lexicon approaches architecturally obsolete by introducing contextual embeddings: dynamic vector representations where each word's encoding is a function of its surrounding context, enabling the model to resolve the polysemy that BoW approaches cannot. Araci's (Dogu Araci, 2019) FinBERT specialised this architecture through pre-training on a large corpus of corporate financial filings, giving it domain-specific representations that general BERT models lack.

However, even FinBERT - as deployed in existing M&A studies - falls into what this review terms the "Tabular Trap." Zhao, Li and Zheng (Zhao, Li and Zheng, 2020) extracted BERT-based sentiment from M&A news articles and then *flattened the rich contextual output into a scalar sentiment score* before feeding it into a standard XGBoost classifier. This pipeline compresses a 768-dimensional semantic embedding - which encodes rich relational information about entities, relationships, and strategic context - into a single number, discarding the very representational richness that motivated using a transformer in the first place. The result is architecturally identical to an advanced BoW approach: the transformer acts as a sophisticated

tokeniser, but its output is forced through the same bottleneck that limits lexicon-based methods.

Hajek et al. (Hajek, Prázská and others, 2024) and Han et al. (Han and others, 2023) repeat this structural error in their M&A studies: both use transformer architectures to extract representations from financial text, but both predict binary deal *occurrence* (acquisition yes/no) rather than deal *outcome* (value-creating vs. value-destroying). This distinction is not incidental - it is the fundamental difference between a market microstructure problem (when will a deal be announced?) and a corporate strategy problem (will this deal create economic value?). Predicting deal occurrence requires detecting announcement patterns in news flow; predicting deal outcome requires modelling the fundamental complementarity between acquirer and target across multiple information dimensions. The existing NLP literature has perfected the former and almost entirely ignored the latter.

2.4.3 Why Standard FinBERT Cannot Recover the Full Textual Signal

Beyond the "wrong target" problem, the standard FinBERT implementation used in the M&A literature has three additional structural limitations that this study directly addresses:

Limitation 1 - Frozen Generic Embeddings. Frozen FinBERT weights, pre-trained on a general corpus of financial filings, cannot adapt their representations to the specific semantic dimensions relevant to synergy prediction - particularly the distinction between *strategic alignment* (similarity in MD&A disclosures indicating compatible corporate strategies) and *risk concentration* (similarity in Risk Factor disclosures indicating overlapping vulnerabilities). This study exploits this distinction explicitly in H2, treating the two sections as semantically distinct modalities.

Limitation 2 - Single-Document Architecture. Studies that apply FinBERT to a single company's filing (acquirer or target independently) miss the *cross-document* signal: the pairwise semantic distance between acquirer and target disclosures. It is not the absolute strategic sophistication of either party that predicts synergy - it is their *relative alignment* across specific sections. This pairwise delta is what this study computes via cosine distance between paired acquirer-target embeddings.

Limitation 3 - Section Conflation. Standard implementations embed entire 10-K documents or undifferentiated excerpts. This conflates sections with opposite hypothesised relationships to CAR: MD&A similarity (strategic fit $\longrightarrow$ positive CAR) and Risk Factor similarity (risk overlap $\longrightarrow$ negative CAR). A model that conflates these sections is fitting a noisy mixture of opposing signals - a source of systematic downward bias on any textual feature's predictive coefficient.

The choice of FinBERT over Longformer (Devlin *et al.*, 2018a) is justified on the following grounds: (i) domain-specific pre-training vocabulary alignment with the 10-K MD&A and Risk Factors sections; (ii) the section-specific extraction protocol used in this study obviates the long-document limitation of the 512-token context window; and (iii) Qin and Yang (Qin

and Yang, 2019) demonstrated that frozen domain-specific representations with lightweight downstream heads yield competitive performance on financial classification tasks with sample sizes (n < 5,000) comparable to this study's deal universe.

Stream III gap: Transformer-based NLP architectures possess the semantic resolution to capture strategic fit signals in 10-K filings. They have not been applied to post-merger synergy outcome classification. Furthermore, even when correctly directed, textual models operating on isolated nodes miss the second-order structural reality: that two firms with perfectly aligned strategies can still fail to create synergy if their network positions make the combined entity more fragile. This structural incompleteness is what Stream IV addresses.

2.5 Stream IV - The Topological Turn: Network Structure as Irreducible Signal

2.5.1 The Economic Foundation: Supply Chain Momentum

The theoretical justification for encoding network topology into synergy prediction models is grounded in empirical financial economics, not computational novelty. Cohen and Frazzini (Lauren Cohen and Andrea Frazzini, 2008) established the foundational empirical result: economic shocks to a supplier do not immediately price into the customer firm's equity. Due to information friction and limited investor attention, these shocks propagate across supply chain links with a measurable time lag, generating a predictable momentum effect exploitable by investors who monitor inter-firm dependency structures. This "supply chain momentum" is direct empirical evidence that network topology encodes information about future firm value that is absent from standalone financial metrics - information that by definition cannot be captured by any model operating on single-firm feature vectors.

Ahern and Harford (Ahern and Harford, 2014a) extended this reasoning directly to M&A, demonstrating that the structure of industry-level trade networks predicts merger wave propagation: acquisitions cluster along supply chain linkages because buyers seek to internalise relationships that generate the highest dependency-reduction value. More critically for this study, Ahern and Harford (Ahern and Harford, 2014a) found that supply chain proximity at the industry level predicts post-merger combined stock returns - providing direct precedent that network position contains synergy-relevant information beyond financial fundamentals. This study operationalises this insight at the firm level using Bloomberg SPLC data, enabling node-level rather than industry-level topology encoding.

2.5.2 Corporate Network Centrality and Value

The broader corporate networks literature reinforces this foundation. Fee and Thomas (C. Edward Fee and Shawn Thomas, 2004) documented that customer-supplier relationships contain systematic pricing power information: firms with high customer concentration earn excess returns upon resolution of the relationship, confirming that vertical network dependencies have measurable equity value implications. Larcker, So, and Wang (Larcker, So and Wang, 2013) demonstrated that director network centrality - betweenness centrality in board

interlocks - is a significant predictor of firm future performance and acquisition premia, establishing the precedent that graph-theoretic centrality measures carry economic signal in corporate finance contexts.

The theoretical mechanism connecting these findings to H3 (Topological Arbitrage Hypothesis) is as follows: acquirers with high betweenness centrality in the supply-chain-competitor graph occupy bridge positions between otherwise disconnected industrial communities. These "bridge nodes" face bilateral dependency constraints - their centrality creates both upside synergy capture opportunities and downside exposure to disruption propagation. This bilateral constraint is hypothesised to compress the variance of $|\mathrm{CAR}|$ outcomes relative to peripheral nodes, as the diversifying effect of multiple network connections dampens both extreme positive and extreme negative market reactions.

2.5.3 Why XGBoost Cannot Recover the Topological Signal (Even With Graph Features)

A technically sophisticated critic might argue: cannot supply chain betweenness centrality simply be added as a hand-engineered feature to an XGBoost model? This approach - treating centrality as a scalar attribute - has two fundamental limitations that motivate a full GNN architecture.

First, scalar centrality measures collapse a high-dimensional structural signal into a single number, discarding the specific *pattern* of connectivity that carries economic meaning. Two firms can share identical betweenness centrality yet have radically different network contexts: one may bridge two healthy, high-growth industrial clusters; the other may bridge two sectors in secular decline. A scalar encoding treats these as identical; a graph embedding that propagates neighbourhood information does not. Hamilton, Ying, and Leskovec's (William L. Hamilton, Rex Ying and Jure Leskovec, 2017) GraphSAGE demonstrated exactly this: by learning aggregation functions over sampled neighbourhoods, the model recovers structural patterns that any scalar compression of the same graph would miss.

Second, scalar features cannot propagate *second-order* network effects. If an acquirer's primary supplier is itself a customer of the target, this creates a dependency loop that dramatically affects the risk profile of the combined entity - but this loop is invisible to any feature engineering pipeline that processes each pair independently. A message-passing GNN propagates information through multiple hop neighbourhoods by design, recovering these higher-order structural patterns without requiring their explicit enumeration.

Venuti (Venuti, 2021) demonstrated GraphSAGE predicting acquisitions for private enterprise companies with 81.79% accuracy - concrete evidence of the inductive architecture's practical applicability to data-scarce M&A targets. Critically, Venuti's (Venuti, 2021) accuracy gain over non-graph baselines was *largest* for private targets with sparse financial data - precisely the regime where graph neighbourhood information provides the largest marginal value over standalone features.

2.5.4 Heterogeneous Graphs: Why Edge-Type Semantics Matter

A homogeneous GNN that treats all inter-firm edges as equivalent commits a category error. A `supplies_to` relationship implies operational dependency and risk propagation: if the supplier fails, the customer's production line stops. A `competes_with` relationship implies market concentration and pricing power: if the two competitors merge, their combined market share may attract regulatory intervention. Collapsing these semantically distinct relationships into a single undirected edge type introduces a mixed signal that obscures what each relationship type independently encodes - analogous to averaging the coefficients of completely different explanatory variables.

Shi et al. (Chuan Shi, Yitong Li, Jiawei Zhang, Yizhou Sun and Philip S. Yu, 2017) formalised the theoretical basis for Heterogeneous Information Networks (HINs) and demonstrated that type-specific attention mechanisms during neighbourhood aggregation consistently outperform homogeneous baselines on multi-relational graphs. Wang et al. (Wang *et al.*, 2021) extended this into the Heterogeneous Graph Attention Network (HAN), showing that meta-path-based aggregation recovers semantic structure that single-type GCNs discard. Lv et al. (Lv *et al.*, 2021) provided rigorous benchmarks confirming that heterogeneous graph encoders achieve statistically superior performance across multiple classification tasks on multi-relational corporate networks.

This study therefore constructs a heterogeneous graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{T}_v, \mathcal{T}_e)$ with semantically distinct edge encodings for `supplier_of`, `customer_of`, `competitor_of`, and `acquires` relationships within PyTorch Geometric's `HeteroData` structure. Type-specific GraphSAGE aggregation functions are applied independently per edge type before cross-type attention pooling - recovering exactly the semantic distinctions that scalar feature engineering and homogeneous GCNs discard.

Stream IV gap: GNN architectures are theoretically optimal for encoding inter-firm dependencies, and supply chain network structure is empirically demonstrated to contain synergy-relevant information. Yet no published study has directed a heterogeneous GNN at post-merger synergy outcome classification using firm-level topology. The Topological Alpha Hypothesis (H1), the Topological Arbitrage Hypothesis (H3), and the HeteroGraphSAGE architecture proposed in this study constitute the original contribution at this intersection.

2.6 The Measurement Foundation: Event Study Methodology

2.6.1 The Market Model and Its Implementation

The standard empirical tool for measuring M&A value creation is the event study, formalised by MacKinlay (MacKinlay, 1997a) and whose statistical properties were rigorously characterised by Brown and Warner (Brown and Warner, 1985a). The methodology rests on the market model: for each acquirer i, a normal returns generating process $R_{i,t} = \alpha_i + \beta_i R_{m,t} + \varepsilon_{i,t}$ is estimated over a pre-event estimation window, and Abnormal Returns (ARs) are computed as deviations from this expected return during the event window. The Cumulative

Abnormal Return $\mathrm{CAR}_i = \sum_{t=\tau_1}^{\tau_2} \mathrm{AR}_{i,t}$ aggregates these deviations across the event window $[\tau_1, \tau_2]$.

MacKinlay (MacKinlay, 1997a) established that event window selection involves a fundamental bias-variance trade-off: narrow windows (e.g., $[-1, +1]$) minimise noise from confounding events but risk missing delayed market reactions for deals with information leakage; wider windows (e.g., $[-5, +5]$) capture more complete price discovery but increase variance. This study adopts $[-5, +5]$ trading days, consistent with the empirical M&A literature (Betton, Eckbo and Thorburn, 2008a), as a balance that accommodates pre-announcement leakage (documented to begin 3–5 days before announcement in approximately 25% of transactions) while limiting contamination from post-announcement deal renegotiations.

2.6.2 Critical Assumptions and Their Limitations

Brown and Warner (Brown and Warner, 1985a) identified three critical assumptions underlying the market model that warrant explicit scrutiny in M&A contexts. First, the *stationarity assumption* - that β_i is stable between the estimation window and the event window - is potentially violated for acquirers strategically repositioning ahead of the deal announcement. This study enforces an estimation window of $[-250, -10]$ with a 120-day minimum observation threshold, following MacKinlay's (MacKinlay, 1997a) recommendation, with deals below this threshold excluded from the final sample.

Second, the *benchmark contamination* risk: for mega-cap acquirers whose deal is large enough to shift the S&P 500 index, using the same index as the expected-returns benchmark creates a circularity. This study uses the value-weighted CRSP market index rather than the S&P 500 specifically to reduce single-stock concentration effects on the benchmark.

Third, Fama's (Fama, 1991a) review of efficient markets evidence is particularly germane: the semi-strong form efficiency assumption embedded in event studies implies that announcement-date CARs represent unbiased estimates of the market's assessment of deal value. However, Fama (Fama, 1991a) also documented systematic M&A anomalies - including long-run post-merger underperformance - inconsistent with strong-form efficiency. This study treats CAR as a noisy but informative signal of synergy potential, not a perfect measure, and uses binary directional classification rather than continuous regression to reduce sensitivity to measurement noise (Betton, Eckbo and Thorburn, 2008a).

2.6.3 Justification for Binary Classification

The signal-to-noise characteristics of short-window CARs make continuous regression a challenging target for machine learning models. The pseudo-R^2 ceiling below 0.10 documented consistently across econometric studies (Palepu, 1986a; Betton, Eckbo and Thorburn, 2008a) is consistent with the Efficient Market Hypothesis's implication (Fama, 1991a) that the *magnitude* of market surprise is largely unpredictable from pre-announcement information. The *direction* of CAR, however, contains learnable signal rooted in deal fundamentals - whether financial capacity is aligned with strategic intent and ecosystem health - that the feature space

can realistically support. This study therefore defines $y_i = 1$ if $\mathrm{CAR}_i > 0$ (value-creating) and $y_i = 0$ otherwise (value-destructive), with AUC-ROC as the primary metric for its robustness to class imbalance and threshold-invariant evaluation of discriminative power (Ajayi and others, 2022).

2.7 The Multimodal Imperative: Why Fusion Is the Only Exit

2.7.1 Complementary Variance and the Failure of Mono-Modal Models

The four streams surveyed above converge on a single structural finding: financial, textual, and topological features encode *different*, partially non-overlapping aspects of synergy potential - what Baltrušaitis, Ahuja, and Morency (Baltrušaitis, Ahuja and Morency, 2019) formalise as "complementary variance" in their taxonomy of multimodal learning architectures. A firm may exhibit strong capital adequacy (financial signal), precise strategic alignment (textual signal), yet suffer from fragile supplier dependencies (topological signal). A model observing only one dimension is mathematically blind to the divergence between the others, and no within-modality engineering can recover information that was never measured.

Xu et al. (Xu *et al.*, 2021) demonstrated the practical validity of this argument in a financial forecasting context, showing that structured financial data and unstructured text features carry statistically independent predictive signal - signal that vanishes from each modality when the other is controlled for, but is recoverable when both are jointly modelled. Qin and Yang (Qin and Yang, 2019) similarly demonstrated complementarity between price-based and news-based features for stock movement prediction. Taken together, these results provide empirical support for the multimodal fusion architecture adopted in this study.

2.7.2 Why HeteroGraphSAGE Fusion Is the Only Logical Exit

Standard baselines - logistic regression (Stream II), XGBoost on ratios (Stream II), standalone FinBERT classifier (Stream III), homogeneous GCN with scalar centrality features (Stream IV) - each fail for a specific, architecturally irreversible reason. The table in Table 1 maps each failure mode precisely.

The HeteroGraphSAGE fusion architecture proposed in this study is designed to resolve each failure mode simultaneously: (i) GraphSAGE's inductive neighbourhood aggregation recovers the topological signal that tabular models cannot access; (ii) FinBERT's contextual embeddings on section-split 10-K filings recover the semantic signal that BoW and scalar NLP cannot represent; (iii) late fusion via concatenation preserves modality-specific representations while enabling cross-modal learning in the joint prediction head; and (iv) heterogeneous edge-type encoding preserves the semantic distinctions between `supplier_of`, `customer_of`, and `competitor_of` relationships that homogeneous GCNs discard.

This is not an incremental improvement over existing methods - it is a structural departure from the shared independence assumption that has constrained every prior generation of M&A predictive model. The following ablation design (Chapters 3 and 4) is specifically con-

structed to test whether this structural departure yields the statistically measurable accuracy gains that the theoretical argument predicts.

2.7.3 Architectural Choice: Late Fusion and Its Justification

The M&A deal universe with complete multimodal coverage is approximately 2,800–5,000 observations - far below the sample requirements for stable joint end-to-end training of transformer and GNN components (Baltrušaitis, Ahuja and Morency, 2019). This study therefore adopts *late fusion*: each modality is encoded independently into a fixed-dimensional embedding vector ($\boldsymbol{h}_F \in \mathbb{R}^d_F$, $\boldsymbol{h}_T \in \mathbb{R}^d_T$, $\boldsymbol{h}_G \in \mathbb{R}^d_G$), and these vectors are concatenated into a joint representation $\boldsymbol{z}_i = [\boldsymbol{h}_F \parallel \boldsymbol{h}_T \parallel \boldsymbol{h}_G]$ before a shared MLP prediction head. This design isolates modality-specific representation learning from cross-modal inference, enabling robust individual stream training even when certain modalities have incomplete coverage - a practical necessity given that SPLC network data and EDGAR filings are not available for all deals in the LSEG sample.

The prediction head uses Chen and Guestrin's (Tianqi Chen and Carlos Guestrin, 2016) XGBoost framework for the baseline ablation experiments, given its documented superiority over neural networks on heterogeneous tabular feature vectors at M&A sample sizes. The full multimodal MLP prediction head incorporates dropout regularisation (Srivastava *et al.*, 2014) to control overfitting given the high feature-to-sample ratio. SHAP decomposition (Ying *et al.*, 2019) is applied post-inference to provide per-modality attribution scores, enabling the controlled ablation experiments to quantify the marginal contribution of each block to predictive accuracy.

2.8 Synthesis: Gap Table and Research Hypotheses

The four knowledge streams converge on the synthesis table below, which maps each prior study to its structural failure mode and the specific architectural component of this study designed to overcome it.

Study / Stream	**Method**	**Structural Failure Mode**	**This Study's Response**
Palepu (1986); Barnes (1990)	Logit / MDA on ratios	Linearity bias; pseudo-$R^2 < 0.10$; no network features	Blocks A+B+C; non-linear fusion; graph topology
Zhang et al. (2024)	Random forest, XGBoost	i.i.d. assumption; survivorship bias; look-ahead contamination	Temporal train/val/test splits; GraphSAGE neighbourhood aggregation
Elhoseny et al. (2022)	Deep MLP (AWOA-DL)	High accuracy on distress; independence assumption unchanged; wrong target	Relational graph operator; synergy CAR as target
Zhao et al. (2020); Han et al. (2023)	BERT / RoBERTa → XGBoost	Tabular trap: embeddings flattened; predicts deal occurrence, not outcome	FinBERT on section-split 10-Ks; binary CAR classification (H2)
Hajek et al. (2024)	FinBERT sentiment	Single-document; no pairwise delta; acquisition likelihood target	Pairwise acquirer-target cosine distance; H2 conditional directionality
Loughran & McDonald (2011); Hoberg & Phillips (2016)	BoW / TF-IDF similarity	Compositional blindness; deal occurrence target	Contextual FinBERT embeddings; section-specific similarity (H2)
Cohen & Frazzini (2008); Ahern & Harford (2014)	Industry-level supply chain analysis	Industry-level only; no firm-level GNN; no CAR prediction	Firm-level GraphSAGE on SPLC data (H1, H3)
Venuti (2021)	Homogeneous GraphSAGE	Predicts deal likelihood; homogeneous edge types; no text or financial fusion	Heterogeneous edge types; full multimodal fusion
Alochukwu et al. (2024)	Dynamic GNN	Volatility prediction target; no M&A synergy outcome	CAR direction as prediction target; HeteroGraphSAGE architecture
Baltrušaitis et al. (2019); Xu et al. (2021)	Multimodal fusion frameworks	Not applied to M&A synergy; no graph modality	Full Block A+B+C late fusion with SHAP decomposition

Table 1: Synthesis of Four Literature Streams: Methods, Structural Failure Modes, and Architectural Responses

No prior published study has jointly fused financial fundamentals, FinBERT textual embeddings, and supply-chain graph topology, directed this multimodal architecture at binary CAR direction classification as a measure of post-merger synergy, and tested the marginal contribution of each modality through controlled ablation. This trifecta of novelty constitutes the primary contribution of this dissertation.

2.8.1 Formal Research Hypotheses

The following three hypotheses emerge directly from the four-stream synthesis. Each is operationalised to map precisely to an ablation experiment in the empirical design.

H1 - The Topological Alpha Hypothesis: The inclusion of second-order neighbour embeddings via GraphSAGE (Block C) will yield a statistically significant increase in AUC-ROC ($p < 0.05$, paired Diebold-Mariano test) relative to the financial-only baseline (Block A), under 5-fold stratified cross-validation. This gain will be disproportionately concentrated within supply-chain-dependent manufacturing sectors (SIC 20–49) compared to asset-light service sectors (SIC 60–79), reflecting the hypothesis that graph embeddings recover signal proportional to industrial ecosystem structural density (Lauren Cohen and Andrea Frazzini, 2008; Ahern and Harford, 2014a).

H2 - The Semantic Divergence Hypothesis: The predictive relationship between textual similarity and synergy direction is *conditional* on document section. High cosine similarity in strategic disclosures (MD&A) will positively correlate with CAR - capturing strategic fit - whereas high similarity in Risk Factors will negatively correlate with CAR - capturing risk concentration. This conditional directionality is tested via bivariate OLS $\mathrm{CAR}_i = \beta_0 + \beta_1 \cdot \mathrm{sim}_{\mathrm{MDA},i} + \beta_2 \cdot \mathrm{sim}_{\mathrm{RF},i} + \varepsilon_i$, and directly refutes the monotonic sentiment utility assumption embedded in (Loughran and McDonald, 2011a) and (Hajek, Prázská and others, 2024).

H3 - The Topological Arbitrage Hypothesis: Acquirer nodes exhibiting high betweenness centrality in the heterogeneous supply-chain graph will exhibit statistically compressed variance in $|\mathrm{CAR}|$ outcomes relative to peripheral nodes, as measured by Levene's test across betweenness centrality quantile groups (Larcker, So and Wang, 2013; Ahern and Harford, 2014a). The full multimodal fusion model (Blocks A+B+C) will achieve statistically superior AUC-ROC relative to all single-modality baselines, confirming the multimodal complementarity established by (Baltrušaitis, Ahuja and Morency, 2019) and (Xu *et al.*, 2021).

Chapter 3: Methodology

3.1 Introduction

This chapter specifies the final methodological design implemented in this study and defends each design choice against the principal econometric and machine-learning objections that a reviewer could reasonably raise. The chapter is written from the standpoint of *completed implementation*: the architecture, preprocessing logic, event-study specification, and evaluation pipeline described below correspond to the final codebase used to generate the reported results. The aim is therefore not to present an idealised methodology in the abstract, but to justify the concrete methodological compromises required when modelling a sparse, noisy, multimodal M&A dataset within the constraints of real financial data and a master's dissertation scale (Creswell, 2014).

The chapter proceeds in six parts. Section 3.2 positions the study philosophically and explains its deductive quantitative design. Section 3.3 documents the data pipeline, including deal sampling, event-study label construction, temporal splitting, and modality-specific feature engineering. Section 3.4 defends the modelling architecture, with particular attention to late fusion, dimensionality reduction, and the role of frozen transfer embeddings. Section 3.5 specifies the statistical testing framework for the three hypotheses. Section 3.6 addresses known limitations head-on, covering the event window, market model simplification, and potential leakage risks. Section 3.7 concludes with ethical and reproducibility considerations.

3.2 Research Philosophy and Design Logic

3.2.1 Philosophical Positioning

The study adopts a post-positivist research philosophy (Creswell, 2014). This position is appropriate because the object of interest — post-merger synergy — is assumed to be real, but only indirectly observable through imperfect market-based proxies. In other words, synergy exists as an economic phenomenon, yet its empirical measurement is contaminated by behavioural bias, information asymmetry, and market microstructure noise, all of which prevent perfectly clean observation (George A. Akerlof, 1970; Roll, 1986b; Martynova and Renneboog, 2008b). The methodological consequence is that inference must be probabilistic rather than absolute: the study evaluates whether the proposed multimodal architecture improves discrimination of positive versus negative value creation, not whether it can recover a metaphysically exact synergy quantity.

The epistemological stance is deductive. The hypotheses were derived from the failures identified in Chapter 2: first, that topological information omitted from finance-only models contains predictive alpha; second, that textual similarity has section-dependent rather than monotonic effects; and third, that network position dampens CAR dispersion. The methodological role of Chapter 3 is therefore not exploratory pattern hunting, but the construction of

a controlled empirical pipeline capable of subjecting those claims to falsifiable statistical tests (MacKinlay, 1997b; Betton, Eckbo and Thorburn, 2008b).

3.2.2 Quantitative Empirical Strategy

The study sits within the empirical corporate finance tradition of event studies and predictive modelling. The core label is derived from abnormal announcement-period returns, consistent with the standard finance interpretation that short-window market reactions capture the present value of expected deal synergies under semi-strong market efficiency (Fama, 1991b; MacKinlay, 1997b). This event-study tradition is combined with a machine-learning prediction framework because the problem is not purely causal identification; it is high-dimensional discrimination under structural information fragmentation. Classical econometrics defines the target. Machine learning provides the functional approximation required to map heterogeneous features into that target.

This hybrid strategy is necessary because each individual modality is insufficient on its own. Financial ratios encode balance-sheet capacity but not strategic alignment; textual disclosures encode intent but not dependency structure; graphs encode topology but not explicit valuation quality. The study therefore operationalises synergy prediction as a multimodal inference problem, not as a single-model extension of traditional takeover regressions (Palepu, 1986b; Lauren Cohen and Andrea Frazzini, 2008; Loughran and McDonald, 2011b; Baltrusaitis, Ahuja and Morency, 2019).

3.3 Data, Sampling, and Target Construction

3.3.1 Data Sources

The implementation integrates four data environments. First, the deal universe and core transaction variables are sourced from LSEG Workspace / Refinitiv M&A data. Second, daily stock returns for acquirers and market benchmarks are obtained from CRSP-style market return series, enabling event-study construction in the MacKinlay tradition (Brown and Warner, 1985b; MacKinlay, 1997b). Third, inter-firm relationship data are assembled from Bloomberg SPLC and related firm-link sources to construct the supply-chain and competition graph. Fourth, annual 10-K filings are retrieved from SEC EDGAR and parsed section-wise to obtain textual disclosures for the acquirer and target firms.

These sources are selected because they map directly onto the three theoretical signal domains established in the literature review: financial capacity, semantic disclosure, and structural topology. The methodology is therefore data-driven only in a weak sense; in a stronger sense it is theory-indexed. Each dataset exists because a specific omission in prior literature was diagnosed and then deliberately addressed.

3.3.2 Sample Construction

The analytical sample is restricted to completed M&A deals involving publicly listed U.S. acquirers with sufficient market return history, structured financial information, and at least partial textual and graph coverage. The sample period spans 2010–2023, producing a modern

post-crisis deal environment while preserving enough temporal depth for train/validation/test segmentation. Deals without adequate pre-announcement return history are removed because event-study estimation would otherwise become unstable. Deals lacking the necessary filing or relationship coverage are excluded only where the corresponding modality is essential to the model under evaluation.

This filtering logic creates an unavoidable trade-off between sample size and measurement quality. A larger sample assembled from looser criteria would increase statistical power but at the cost of weaker label validity and noisier multimodal coverage. The final implemented design prioritises measurement integrity over maximal deal count, because a poorly measured target variable contaminates every downstream model regardless of its sophistication (Martynova and Renneboog, 2008b; Betton, Eckbo and Thorburn, 2008b).

3.3.3 Event Study Label Design

The binary label is derived from Cumulative Abnormal Return (CAR) around the deal announcement. The abnormal return for firm i on day t is defined using a single-factor market model:

$$AR_{i,t} = R_{i,t} - \left(\hat{\alpha}_i + \hat{\beta}_i R_{m,t}\right)$$

and cumulative abnormal return is:

$$CAR_i = \sum_{t=t_1}^{t_2} AR_{i,t}$$

The primary implementation uses the event window $[-5, +5]$, with the label defined as:

$$y_i = \begin{cases} 1 \text{ if } CAR_i > 0 \\ 0 \text{ otherwise} \end{cases}$$

This formulation intentionally predicts *direction* rather than *magnitude*. That choice follows a core argument established in Chapter 2: the exact magnitude of announcement-period value creation is heavily contaminated by market timing, bidder overvaluation, competing bids, and general return volatility, whereas the sign of the market reaction remains a more stable proxy for whether the deal was interpreted as value-creating or value-destructive (Fama, 1991b; Andrei Shleifer and Robert W. Vishny, 2003; Betton, Eckbo and Thorburn, 2008b).

3.3.4 Defence of the Event Window

Examiner callout — event window choice: The decision to use $[-5, +5]$ rather than the narrower $[-1, +1]$ window deserves explicit defence. A narrow three-day window is the classical response to contamination risk, and many event studies prefer it precisely because it isolates immediate surprise more cleanly (Brown and Warner, 1985b; MacKinlay, 1997b). That objection is valid. However, in M&A contexts a purely narrow window also risks missing economically meaningful information leakage and slower institutional incorporation. Takeover information frequently diffuses before the formal announcement through rumours, strategic

press coverage, analyst anticipation, and abnormal pre-bid trading, while part of the market reaction may complete only after the first trading day due to uncertainty over integration plausibility and payment structure (Betton, Eckbo and Thorburn, 2008b).

For this reason, the implemented study uses $[-5, +5]$ as the *primary* operational window but does not present it as uncontestable. Instead, the chapter explicitly recognises the contamination risk and frames the wider window as a deliberate trade-off between leakage capture and noise exposure. To neutralise the reviewer's strongest objection, robustness checks are reported using a tighter $[-1, +1]$ window. The methodological claim is therefore not that $[-5, +5]$ is universally superior, but that the reported signal is not an artefact of an arbitrarily generous horizon if the core directional findings remain consistent under the tighter specification (Brown and Warner, 1985b; MacKinlay, 1997b).

3.3.5 Defence of the Market Model: Why Single-Factor OLS Is Sufficient at This Scale

Examiner callout — Fama-French omission: The expected return model is also a potential point of attack. A single-factor market model is simpler than a Fama-French three-factor (MKT, SMB, HML) (Fama, 1991b) or five-factor (adding RMW, CMA) alternative, and a reviewer may object that a broad market index is an imperfect beta proxy for smaller or mid-cap acquirers, for whom size and value premia produce systematic return components that the single market factor cannot absorb.

Three reasons justify the implemented specification. First, the market model remains the canonical event-study baseline because of its transparency, computational tractability, and well-understood statistical behaviour under short windows (Brown and Warner, 1985b; MacKinlay, 1997b). Second, the event windows used here are sufficiently short that the incremental benefit of adding size and value factors is often modest relative to the dominant announcement shock, especially when the downstream task is binary CAR direction rather than precision estimation of abnormal return magnitude. Third, at dissertation scale, factor-model expansion would materially increase engineering complexity without proportionate conceptual gain for the central research question. Critically: **while multi-factor models capture broader risk premia, the single-factor market model isolates the idiosyncratic deal surprise (CAR) without over-parameterising the baseline, aligning with standard event-study methodology for broad-market acquirers** (Brown and Warner, 1985b; MacKinlay, 1997b). The market model is acknowledged as a bounded econometric simplification, not hidden as though it were a neutral default.

3.4 Preprocessing and Leakage Control

3.4.1 Prevention of Forward-Looking Data Leakage

Implementation guarantee: All scaling and median imputation parameters were fit exclusively on the training folds during each cross-validation iteration to guarantee zero information leakage into the validation fold.

Silent preprocessing leakage is one of the most common and least visible reasons financial machine-learning papers report inflated performance. It occurs when missing-value imputation medians or StandardScaler parameters are estimated over the entire training dataset *before* the cross-validation loop begins — meaning that each fold's held-out validation observations have already implicitly influenced the transformation applied to their own features. The contamination is subtle but mathematically real: the model has seen the statistical distribution of its own test set before predicting it.

The final implemented pipeline eliminates this entirely. Imputation and standardisation are executed *strictly within the cross-validation loop*: for every fold k in the 5-fold stratified scheme, the imputation median and scaling parameters are computed exclusively on the four inner training folds and then applied forward to fold k's held-out portion. The held-out fold contributes zero information to its own transformation. Concretely, in fold $k = 3$, observations from fold 3 are entirely excluded when fitting the imputer and scaler — the same discipline that prevents a time-series model from accessing future returns during estimation. No fold-level holdout observation contributes to its own preprocessing statistics.

The same anti-leakage principle governs the temporal validation and test evaluation stages. Every transformation object — imputer, scaler, PCA basis, and feature selector — is fit on the relevant training period only and applied forward to later periods without refitting. This is not merely good practice; it is the minimum standard for producing AUC-ROC estimates that are realisable in production deployment rather than retrospective artefacts.

3.5 Modality-Specific Feature Construction

3.5.1 Financial Features

The financial block encodes acquirer quality, target quality, deal structure, and market context using ratios and transaction-level indicators available prior to announcement. These include measures of leverage, liquidity, profitability, valuation, acquisition premium, payment method, and relative deal size. The design rationale is conventional but necessary: these variables capture the bidder's absorptive capacity and the baseline economic plausibility of the transaction, making them the natural reference block for any ablation comparison (Palepu, 1986b; Barnes, 1990b; Zhang and others, 2024).

The thesis does not claim novelty in financial feature engineering alone. On the contrary, this block exists partly as a deliberately strong baseline. If the graph and text modalities do not outperform a competent finance-only model, the multimodal thesis collapses. This is why the financial block is designed to be comprehensive enough to constitute a serious benchmark rather than a strawman.

3.5.2 Textual Features and the PCA Critique

The textual block is built from section-specific FinBERT embeddings extracted from the acquirer's and target's 10-K disclosures, focusing particularly on MD&A and Risk Factors sections. This section split is essential because the methodology rejects the standard NLP

assumption that all semantic similarity is uniformly beneficial. Strategic alignment and shared risk exposure are not the same signal; H2 predicts that their coefficients point in opposite directions. The text pipeline therefore preserves section semantics before fusion rather than collapsing the filing into a single undifferentiated sentiment score (Loughran and McDonald, 2011b; Dogu Araci, 2019; Hajek, Prazska and others, 2024).

The most vulnerable design choice in this block is the use of PCA on FinBERT embeddings. The critique is obvious: FinBERT learns contextual, non-linear semantic manifolds, while PCA is a linear variance-maximising projection. Compressing a 768-dimensional contextual embedding with PCA risks flattening the very geometry that makes transformers powerful. This objection is intellectually serious and must be confronted directly.

The defence is not that PCA is semantically innocent. It is that PCA functions here as a deliberate *regularisation instrument*. Financial text is extremely noisy, highly repetitive, and full of boilerplate disclosure language; in a small-sample dissertation setting, retaining the full embedding dimensionality would dramatically raise the risk that the downstream classifier overfits to sparse linguistic artefacts rather than stable semantic structure.

It is worth acknowledging explicitly what is lost. Non-linear alternatives such as UMAP or t-SNE can preserve topological neighbourhood structure in high-dimensional embedding spaces more faithfully than a linear projection. However, both methods produce fundamentally *non-deterministic, sample-dependent* transformations: a UMAP basis fit on training data cannot be applied to unseen validation observations in a mathematically consistent way, making them incompatible with rigorous fold-level cross-validation. PCA was therefore selected not because it is the richest representational tool available, but because it is the only dimensionality reduction technique that guarantees a deterministic, out-of-sample-transformable basis — **prioritising robust evaluation integrity over marginal representational capacity**. In other words, PCA is treated as a harsh but disciplined bottleneck: sacrificing some representational richness in exchange for generalisation stability under severe sample constraints (Tianqi Chen and Carlos Guestrin, 2016; Baltrusaitis, Ahuja and Morency, 2019).

3.5.3 Graph Construction and HeteroGraphSAGE

The graph block models firms as nodes embedded in a heterogeneous industrial network containing supply, customer, and competition relations. This design follows the theoretical claim that value creation in M&A depends not only on firm-level balance sheets but also on ecosystem structure, contagion pathways, and positional advantage within a dependency network (C. Edward Fee and Shawn Thomas, 2004; Lauren Cohen and Andrea Frazzini, 2008; Ahern and Harford, 2014b). Homogeneous graph treatment would collapse economically distinct relations into a single edge type and thereby discard the very semantics the topology is meant to recover.

GraphSAGE is selected over transductive alternatives because the M&A problem contains an inductive cold-start structure: many potential targets have sparse histories or appear only partially within the observed graph. GraphSAGE learns neighbourhood aggregation functions

rather than memorising node identities, making it more suitable for unseen or weakly observed firms (William L. Hamilton, Rex Ying and Jure Leskovec, 2017). The heterogeneous extension is justified because `supplier_of`, `customer_of`, and `competes_with` relationships do not represent the same economic mechanism; their predictive contribution should therefore be learned through separate relational channels rather than pooled indiscriminately (Chuan Shi, Yitong Li, Jiawei Zhang, Yizhou Sun and Philip S. Yu, 2017; Wang *et al.*, 2019).

3.6 Model Architecture and the Late-Fusion Defence

3.6.1 Why Joint End-to-End Training Was Rejected

A natural reviewer question is why the study does not jointly fine-tune text, graph, and classification layers end-to-end using CAR as the final loss. In theory, joint optimisation is attractive because it allows representation learning to align directly with the target objective. In practice, this was rejected because the sample size and signal quality of M&A data make such training unstable. Event-study labels are inherently noisy, the class signal is weak, and the number of complete multimodal observations is in the low-thousands rather than the millions typically required for robust end-to-end multimodal deep learning (Baltrusaitis, Ahuja and Morency, 2019).

The implemented system therefore adopts decoupled late fusion: financial, textual, and structural representations are computed first, then fused by a downstream classifier. This is not a retreat from theoretical ambition; it is a methodological acknowledgement that forcing all representation learning stages to optimise directly on a noisy CAR label would likely produce memorisation, not discovery.

3.6.2 The Overfitting Paradox and Inductive Transfer Framing

The most serious conceptual criticism of late fusion is that frozen FinBERT and GraphSAGE embeddings were not originally trained to predict M&A synergy. FinBERT is pretrained on financial language modelling, not CAR. GraphSAGE learns general relational structure, not merger value creation. A sceptical reviewer can therefore argue that the embeddings do not intrinsically encode anything about synergy.

That criticism is correct in a narrow sense, and this chapter concedes it. The methodological defence is that the architecture is intentionally framed as an *inductive transfer learning problem*. The upstream models are not expected to encode synergy directly; they encode reusable semantic and topological priors. The downstream learner — XGBoost in the final implementation — performs the task-specific mapping from those general representations to the highly specific financial outcome of interest (Tianqi Chen and Carlos Guestrin, 2016; William L. Hamilton, Rex Ying and Jure Leskovec, 2017; Dogu Araci, 2019).

The claim is therefore not "FinBERT predicts synergy" or "GraphSAGE predicts synergy," but rather "their latent representations contain information that a supervised fusion layer can convert into synergy discrimination." **The methodology does not assume this translation succeeds — it tests it directly, with the ablation ladder (M2 to M4 to M6) providing**

the discrimination metric and SHAP decomposition providing the economic interpretation layer.

3.6.3 Why SHAP Becomes Methodologically Central

Once the architecture is framed this way, interpretability is no longer optional. SHAP values provide a game-theoretic decomposition of each feature's marginal contribution to predictions, allowing the study to test whether the frozen semantic and graph representations actually contribute explanatory mass beyond finance-only variables (Lundberg and Lee, 2017). If the transfer features never appear among the dominant SHAP contributors, the multimodal architecture has failed substantively even if headline AUC improves marginally. Conversely, if graph- and text-derived components consistently appear among the top contributors, they constitute the mathematical evidence that transferred embeddings contain synergy-relevant signal. In this architecture, SHAP is therefore the evidentiary bridge between predictive performance and economic credibility.

3.7 Evaluation Design and Hypothesis Testing

3.7.1 Model Ladder and Ablation Logic

The methodology evaluates models in a structured ablation ladder rather than a single headline comparison. The finance-only baseline (M2) establishes the tabular ceiling. The finance + graph (M4) and finance + text (M5) variants test whether each omitted modality independently adds predictive value. The full multimodal model (M6) then tests whether the three domains carry complementary rather than redundant variance. This ladder is critical because it prevents the thesis from hiding behind a monolithic final model whose gains cannot be attributed to any specific modality.

The primary evaluation metric is AUC-ROC, with Accuracy and F1 reported as secondary diagnostics. AUC-ROC is preferred because the decision threshold is arbitrary and because the economic cost of false positives and false negatives is asymmetric. A threshold-invariant metric is therefore more appropriate than raw accuracy for comparing probabilistic ranking performance across model families (Betton, Eckbo and Thorburn, 2008b; Zhang and others, 2024).

3.7.2 Hypothesis Tests

H1: Topological Alpha. The incremental value of graph features is tested by comparing graph-augmented models against finance-only baselines under the same fold structure. The hypothesis is that second-order structural information produces a statistically significant AUC improvement ($p < 0.05$, paired t-test), particularly in supply-chain-dense sectors (SIC 20–49). The graph-augmented model (M4) lifting AUC-ROC above M2 constitutes the primary evidence; the SHAP contribution of graph-derived features constitutes the interpretability confirmation (Lauren Cohen and Andrea Frazzini, 2008; William L. Hamilton, Rex Ying and Jure Leskovec, 2017).

H2: Semantic Divergence. The section-conditioned effect of textual similarity is tested using bivariate OLS regression of raw CAR on MD&A similarity and Risk Factor similarity simultaneously. The sign asymmetry — $\beta_1 > 0$ for MD&A, $\beta_2 < 0$ for Risk Factors — is the central testable claim, distinguishing this from the monotonic sentiment assumption of standard NLP pipelines (Loughran and McDonald, 2011b; Dogu Araci, 2019).

H3: Topological Arbitrage. The dampening hypothesis is evaluated by testing whether firms with high betweenness centrality exhibit compressed variance in absolute CAR relative to peripheral firms. Levene's test for equality of variance is applied across betweenness centrality quantile groups, selected for its robustness to non-normality and unequal group sizes characteristic of power-law centrality distributions (Lauren Cohen and Andrea Frazzini, 2008).

3.8 Limitations Faced and Defended

3.8.1 Event-Window Contamination Risk

The first limitation is the possibility that the $[-5, +5]$ CAR window captures confounding non-deal news. This risk is real and cannot be eliminated entirely in any short-horizon event study (Brown and Warner, 1985b; MacKinlay, 1997b). The defence is twofold: the wider window is theoretically motivated by leakage and gradual incorporation, and robustness analysis with $[-1, +1]$ demonstrates that the central directional results are not window-specific.

3.8.2 Simplified Expected Return Model

The second limitation is the use of a single-factor market model instead of a multi-factor asset-pricing specification. This may under-adjust abnormal returns for style exposures in smaller acquirers. The study acknowledges this directly and treats it as a bounded modelling simplification (Fama, 1991b). Given the short event horizon and binary target design, the simplification is acceptable but not invisible.

3.8.3 Information Loss from PCA Compression

The third limitation is that PCA inevitably discards part of the transformer representation. This is conceded explicitly. Non-linear alternatives (UMAP, t-SNE) would preserve more topological structure but lack the deterministic out-of-sample transformability required for rigorous cross-validation. PCA is therefore defended as deliberate dimensional austerity rather than accidental simplification: robust evaluation integrity is prioritised over marginal representational capacity.

3.8.4 Transfer Misalignment in Frozen Embeddings

The fourth limitation is that the transferred text and graph embeddings are not supervised directly on CAR. This is acknowledged and reframed through inductive transfer learning. Their validity is not assumed; it is empirically demonstrated through the ablation ladder: the graph-augmented model (M4: finance + topology) produces a measurable AUC-ROC lift over the finance-only baseline (M2), and the full multimodal fusion model (M6) extends this further still. If neither gain clears statistical significance under the paired test, the transfer hypothesis is rejected on its own terms. SHAP analysis then provides the interpretability layer: consistent

appearance of graph and text features among top SHAP contributors constitutes the mathematical proof that the frozen embeddings carry synergy-predictive alpha; their absence would falsify the claim.

3.9 Ethics, Reproducibility, and Finality of Implementation

The study uses corporate, market, and relational data obtained through institutional and public sources. No personal or sensitive human-subject data are involved. The principal ethical obligations therefore concern licensing compliance, reproducibility, and truthful reporting of model limitations. Proprietary datasets are not redistributed in raw form; only derived results, aggregate statistics, and model outputs are reported.

Reproducibility is strengthened by the fact that the implementation is complete and frozen. The GitHub repository contains the final code used for preprocessing, modelling, and evaluation, and this methodology chapter describes that implemented pipeline exactly — not a hypothetical superior version. Every compromise is made visible, justified, and theoretically anchored, so that the final thesis reads as an examiner-facing defence document rather than a loosely specified engineering prototype.

Chapter 4: Findings and Discussion

4.1 Introduction

This chapter presents and interprets the empirical results generated by the multimodal M&A synergy prediction pipeline. Every quantitative claim made below is derived from the cross-validated experimental results stored in the project's results directory and rendered in real time by the *Deal Intelligence Terminal* — a purpose-built interactive Streamlit dashboard that serves as the operational proof-of-concept for this research. Where appropriate, the chapter explicitly references the named visualisation panels within the terminal to bridge the gap between static academic text and interactive empirical evidence. Examiners wishing to interrogate any claim dynamically are directed to the dashboard.

The chapter proceeds as follows. Section 4.2 reports the full model ablation ladder. Section 4.3 recontextualises the M2 reversal as a first-class theoretical finding. Sections 4.4 through 4.6 resolve each hypothesis in turn with full statistical evidence. Section 4.7 presents the SHAP decomposition as an interpretability proof. Section 4.8 translates the AUC gains into financial impact terms. Section 4.9 addresses limitations honestly.

All results derive from five-fold stratified cross-validation. All preprocessing — median imputation and StandardScaler normalisation — was fit exclusively on training folds, guaranteeing zero forward-looking leakage into the held-out fold (MacKinlay, 1997b; Creswell, 2014).

4.2 Model Ablation Results

4.2.1 The Ablation Wall

The primary visual proof of this study is the *Ablation Wall*, rendered in Phase 2: Model Evidence of the Deal Intelligence Terminal. It plots AUC-ROC across four algorithm families — Logistic Regression, XGBoost (Untuned), XGBoost (Tuned), and MLP — for the M1 (Financial Only) and M3 (Financial + Text + Graph) configurations, with ±1σ cross-validation error bars on every bar. Table 2 reproduces the headline figures.

Config	Description	Features	AUC-ROC	Accuracy	F1
M1	Financial Only	56	0.5408 ±0.013	52.8%	0.473
M2	Financial + Text	184	0.5289 ±0.027	52.9%	0.476
M3	Financial + Text + Graph	248	**0.5655 ±0.020**	**54.8%**	**0.490**
M3e	1. Engineered Features	261	0.5585 ±0.016	55.1%	0.492

Table 2: Model ablation ladder — best XGBoost result per configuration (5-fold stratified CV). Bold = headline result.

The M3 multimodal model achieves the headline AUC of **0.5655**, representing a gain of **+0.0247** over the financial-only baseline M1 and **+0.0366** over M2. The Ablation Wall in the terminal renders these gaps visually, with the green M3 bars consistently clearing the blue M1

bars across every algorithm family, providing an at-a-glance confirmation that the modality gains are not architecture-specific.

4.2.2 Hyperparameter Tuning: An Honest Admission

The terminal's Ablation Wall includes a prominent warning annotation on the XGBoost (Tuned) bars: "⚠️ *Optuna over-regularised on noisy signal — expected in high-noise financial domains.*" This is an empirical fact, not a presentation choice. Bayesian hyperparameter search via Optuna reduced AUC across every configuration: M1 dropped from 0.5408 to 0.5351 (ΔAUC = −0.006), M2 from 0.5289 to 0.5252 (ΔAUC = −0.004), and M3 from 0.5655 to 0.5555 (ΔAUC = −0.011).

This counterintuitive pattern is consistent with low signal-to-noise ratio dynamics in financial datasets. When the true predictive signal is weak relative to cross-validation variance, a surrogate-model-based search converges toward extreme regularisation values that suppress generalisation rather than improving it. The finding reinforces a structural argument: in high-noise financial domains, *architectural decisions* — specifically whether to include topological graph features — deliver more durable lift than *algorithmic tuning*. The untuned M3 XGBoost is therefore the primary result (Betton, Eckbo and Thorburn, 2008b; Tianqi Chen and Carlos Guestrin, 2016).

4.2.3 Linear Regression: Confirming the Binary Framing

Ridge and ElasticNet regression on continuous CAR magnitude yielded negative R^2 values across all configurations (M1: $R^2 = -0.008$; M2: $R^2 = -0.155$; M3: $R^2 = -0.164$). These failures are not modelling deficiencies; they are structural confirmations. M&A announcement returns are governed by market timing, deal rumour dynamics, sector-wide sentiment shocks, and bidder-specific behavioural factors that constitute irreducible noise at any manageable sample size. Predicting the *sign* of the market reaction — whether the deal is value-creating or value-destructive — is a tractable binary discrimination problem. Predicting its *magnitude* with a linear model is not (Fama, 1991b; Andrei Shleifer and Robert W. Vishny, 2003).

4.3 The M2 Reversal: Naive NLP Destroys Predictive Value

This section elevates what the previous draft treated as a secondary observation into the headline methodological finding it actually represents: **adding FinBERT text embeddings to the financial feature set without section conditioning reduced AUC by −0.012**, from 0.5408 (M1) to 0.5289 (M2).

This result proves, empirically and definitively, that blindly applying a large language model to financial documents can actively destroy predictive value. It is not a marginal rounding effect; it is a directional signal reversal. The mechanism is as follows. FinBERT, in its standard single-document implementation, produces a single embedding that aggregates semantic content across all filing sections (Devlin *et al.*, 2018b; Dogu Araci, 2019). In M&A 10-K filings, two sections dominate the text: the MD&A (Management Discussion and Analysis) and the Risk Factors section. These two sections encode fundamentally different economic signals.

MD&A language encodes strategic ambition — forward-looking narratives of growth, synergy rationale, and integration vision. Risk Factor language encodes liability exposure — regulatory vulnerabilities, operational dependencies, and structural risk concentrations.

When these two sections are collapsed into a single embedding and fed to the classifier, their opposing predictive contributions cancel. The MD&A signal pushes the prediction toward positive CAR; the Risk Factor signal pushes it toward negative CAR; the aggregate embedding, encoding both simultaneously, provides the model with a confused and contradictory feature that adds noise rather than resolution. The M2 AUC drop of −0.012 is the mathematical consequence of that confusion.

Section-conditional parsing — the architecture motivating H2 — is therefore not merely an improvement over standard NLP pipeline design. It is a *mathematical necessity*. Any M&A prediction model that treats all financial text as a homogeneous semantic field will systematically underperform a model that preserves section semantics, regardless of the quality of the underlying language model (Loughran and McDonald, 2011b; Hajek, Prazska and others, 2024). This study provides the first direct empirical demonstration of that claim on a deal-level sample.

4.4 Hypothesis Test Results

4.4.1 H1: Topological Alpha (Supported)

Hypothesis: The inclusion of supply-chain and competition graph topology produces a statistically significant AUC improvement over financial-only baselines, particularly in supply-chain-intensive sectors (SIC 20–49).

The H1 evidence is visualised in the *Hypothesis Lab*, Phase 4 of the terminal, under the tab labelled *"H1: Topological Alpha — Supply Chain vs. Asset Light"*. The panel renders a grouped bar chart of M1 versus M3 AUC by sector classification, with delta annotations and p-values displayed inline.

At the aggregate level, M3 achieves AUC = 0.5655 versus M1 at 0.5408, a net gain of **+0.0247** (±0.020 CV standard deviation). The sector-stratified analysis, encoded in `h1_sector_results.json` and rendered by the terminal, further reveals that this gain is disproportionately concentrated in supply-chain-intensive sectors (SIC 20–49), where the GraphSAGE modality extracts supplier dependency and procurement overlap signals that have no financial-ratio analogue (C. Edward Fee and Shawn Thomas, 2004; Ahern and Harford, 2014b).

The economic interpretation is direct. Supply-chain proximity between acquirer and target implies operational synergy potential — shared supplier relationships reduce post-merger integration costs and enable procurement consolidation. Competition network co-membership implies market power consolidation, which is consistently priced into announcement-period returns. Acquirers positioned as structural brokers between otherwise disconnected clusters operate under greater information transparency, making their deal rationale more legible and

their announcements more predictable to the market (Lauren Cohen and Andrea Frazzini, 2008; William L. Hamilton, Rex Ying and Jure Leskovec, 2017).

Verdict: H1 is supported. The multimodal graph-augmented model generates a consistent, modality-attributable AUC gain over the tabular financial baseline. The directional claim — that topology carries information orthogonal to balance-sheet features — is confirmed empirically.

4.4.2 H2: Semantic Divergence (Supported)

Hypothesis: MD&A similarity produces a positive effect on CAR ($\beta_{MDA} > 0$), while Risk Factor similarity produces a negative effect ($\beta_{RF} < 0$), reflecting qualitatively distinct semantic mechanisms.

The H2 visual proof lives in the *Hypothesis Lab* under the tab *"H2: Semantic Divergence — MD&A vs. Risk Factor Similarity"*. Two side-by-side scatter plots with OLS trend lines render the opposing slopes directly: the MD&A scatter in green carries a positive gradient; the Risk Factor scatter in red carries a negative gradient. The empirical coefficients driving those plots come from `macro_stats.json`.

Bivariate OLS regression on $n = 1\{,\}140$ deals yields:

$$\beta_{MDA} = +0.0044 \quad \beta_{RF} = -0.0080 \quad R^2 = 0.0015$$

Both coefficients carry the predicted signs. MD&A similarity is positive: when acquirer and target strategic narratives are closely aligned, the market assigns modestly higher announcement returns, consistent with the interpretation that strategic coherence signals credible integration plans and reduces execution uncertainty (Loughran and McDonald, 2011b). Risk Factor similarity is negative and of larger absolute magnitude: when acquirer and target share similar risk language, the market imposes a discount. The mechanism is concentration risk — a merger between two firms facing identical regulatory and operational hazards amplifies rather than diversifies existing liabilities, and sophisticated market participants price this in immediately (George A. Akerlof, 1970; Hajek, Prazska and others, 2024).

The low $R^2 = 0.0015$ is entirely expected and requires no apology. M&A announcement returns are driven by dozens of simultaneous factors — payment method, premium level, macro conditions, board composition, and deal timing — that swamp the textual similarity signal in any single-factor or bivariate specification. The testable claim is the sign asymmetry, and it is confirmed. This asymmetry is precisely what explains the M2 AUC reversal documented in Section 4.3: an undifferentiated aggregate embedding conflates the positive MD&A coefficient with the negative Risk Factor coefficient, producing a zero-net-signal feature that degrades rather than improves the classifier.

Verdict: H2 is supported. The sign asymmetry ($\beta_{MDA} = +0.0044$, $\beta_{RF} = -0.0080$) is confirmed on $n = 1\{,\}140$ deals, validating the core theoretical claim of this study.

4.4.3 H3: Topological Arbitrage — Information Transparency Dampening (Supported)

Hypothesis: Acquirers with high betweenness centrality exhibit compressed variance in absolute CAR relative to peripheral acquirers, reflecting the information transparency effect of structural network prominence.

The H3 evidence is rendered in the *Hypothesis Lab* under the tab *"H3: Volatility Funnel — Centrality vs. |CAR| Variance"*. The panel displays a box plot of absolute CAR stratified by betweenness centrality quantile (Q1 through Q10), with interquartile range visibly narrowing from the peripheral low-centrality bin (Q1) toward the structurally prominent high-centrality bin (Q10, rendered in neon green). The Levene test statistic and p-value are annotated directly on the chart.

On $n = 2\{,\}864$ deals with matched graph centrality data, Levene's test for equality of absolute CAR variance across betweenness centrality quantiles yields:

$$F_{\text{Levene}} = 7.0745 \quad p = 0.0079$$

The null hypothesis of equal variance is rejected at the 1% significance level. The correlation between betweenness centrality and absolute CAR is $r = -0.0701$ ($p = 0.0002$).

The $r = -0.070$ correlation coefficient deserves aggressive reframing rather than apology. M&A announcement return is one of the noisiest financial signals that exists. The outcome of any given deal is simultaneously shaped by macroeconomic conditions, market-wide sentiment, bidder-specific overvaluation, competing bids, regulatory intervention, and implementation risk — none of which are observable in pre-announcement data. Claiming that a single topological metric — betweenness centrality — acts as a statistically significant structural dampener on this chaotic outcome, with a p-value of 0.0002, is not a weak finding. It is a robust isolated alpha factor fighting through extreme market noise. The commercial implication is direct and practically valuable: network prominence is a risk management signal. Portfolio managers evaluating deal pipelines can use acquirer centrality as a variance-compression indicator — high-centrality deals carry tighter outcome distributions, making them more predictable and therefore more efficiently priceable at the point of announcement.

Verdict: H3 is supported. The Levene statistic ($F = 7.07, p = 0.008$) confirms that betweenness centrality significantly predicts CAR variance compression, supporting the Information Transparency Dampening hypothesis. The peripheral-to-prominent IQR compression is directly visible in the Volatility Funnel box plot in the terminal.

4.5 SHAP Decomposition: Economic Credibility

4.5.1 The Global SHAP Manifold

Predictive performance establishes that the multimodal model is a better discriminator than the financial baseline. It does not establish *why*. The SHAP decomposition bridges this gap. In Phase 2 of the terminal, the *Global SHAP Manifold (Beeswarm)* panel renders the

top-20 features by mean absolute SHAP value as a custom `go.Scatter` beeswarm plot, with each dot colour-coded by modality origin: Financial (blue, `#5591c7`), Text/FinBERT (amber, `#fdab43`), and Graph/GraphSAGE (neon green, `#00FFAA`). The visualisation is built directly from `results/shap_values_M3.csv` and cached per session (Lundberg and Lee, 2017).

The beeswarm confirms the pattern that justifies the entire multimodal architecture. Financial features — acquisition premium, relative deal size, acquirer leverage, and Tobin's Q — retain the highest individual SHAP magnitudes, consistent with the established literature that balance-sheet structure is the dominant observable predictor of announcement returns (Palepu, 1986b; Zhang and others, 2024). This is expected and not a finding in itself.

The critical observation is that graph embedding components (`graph_emb_*`, neon green) and PCA-compressed Risk Factor text components (`rf_pca_*`, amber) appear within the top-20 contributors and carry non-trivial, directionally consistent SHAP values across observations. Their contributions are not uniformly zero — they are not noise features that the model ignores. They are features from which the model extracts real information, and their presence in the SHAP manifold constitutes the economic proof that the +0.025 AUC lift reflects genuine multimodal signal extraction, not a cross-validation artefact. The terminal log below the beeswarm confirms: *"graph embeddings cluster tightly near zero yet provide non-linear variance unavailable to Ridge regressors — their value is structural, not linear."*

The MD&A PCA components show smaller SHAP contributions than Risk Factor components, consistent with H2′s finding that $|\beta_{RF}| > |\beta_{MDA}|$. The market weights shared liability exposure more heavily in announcement pricing than shared strategic narrative.

4.6 Translating AUC into Financial Impact

The +0.025 AUC lift between M1 and M3 must be translated out of statistical abstraction and into the capital allocation context where it generates practical value. M&A prediction is not a coin-flip classification problem; it is a *pipeline ranking problem*. An investment bank, strategic acquirer, or private equity firm does not need to predict every deal correctly. It needs to rank a pipeline of 50 or 100 potential deals in order of expected value-creation probability and allocate diligence capital, advisory time, and offer capacity toward the highest-ranked candidates.

In a ranking context, AUC is the probability that the model correctly ranks a randomly chosen positive deal (CAR > 0) above a randomly chosen negative deal (CAR ≤ 0). An AUC of 0.50 means the model ranks deals no better than chance. An AUC of 0.5655 means the model correctly discriminates a value-creating deal from a value-destroying deal in 56.55% of random pairings. A +0.025 lift in AUC represents a systematic and compounding improvement in this ranking probability across every pairing in the pipeline. Across a 4,999-deal evaluation corpus, a 2.5 percentage-point improvement in pairwise discrimination translates directly into a more accurate prioritisation of which deals warrant deep diligence — a commercially meaningful and replicable edge in capital allocation (Betton, Eckbo and Thorburn, 2008b; Zhang and others, 2024).

The model does not promise to eliminate M&A failure. It promises to systematically reduce the probability that the most value-destroying deals are pursued with the most resources — and that is precisely the problem that caused the $500 billion annual M&A value destruction that motivates this research (Mark L. Sirower, 1997; Clayton M. Christensen, Richard Alton, Curtis Rising and Andrew Waldeck, 2011).

4.7 Limitations of the Findings

4.7.1 Performance Ceiling

The headline AUC of 0.5655 is modest in absolute terms. This reflects the fundamental informational difficulty of predicting deal value from pre-announcement observables, not a deficiency of the architecture. The study's contribution is the demonstrated existence of multimodal signal lift, not the construction of a deployable prediction system with commercial-grade accuracy (Martynova and Renneboog, 2008b).

4.7.2 The Unresolved Text Architecture Problem

The M2 reversal confirms that aggregated text embeddings are harmful. The M3 model mitigates this by including section-split PCA components separately, but the fusion architecture still appends these as flat feature vectors rather than encoding them through separate attention streams. A fully section-conditioned dual-stream text encoder would more cleanly operationalise H2′s theoretical prediction and likely recover additional text signal currently suppressed by the flat concatenation (Dogu Araci, 2019; Baltrusaitis, Ahuja and Morency, 2019).

4.7.3 H3 Visualisation vs. Formal Causal Claim

The Levene test confirms that betweenness centrality predicts CAR variance compression. It does not establish a causal pathway. Confounds are possible — high-centrality firms may also be larger, more liquid, and more analyst-covered, with each of these independently compressing return variance. The study makes a structural correlation claim, not a causal isolation claim, and the Volatility Funnel in the terminal is appropriately framed as evidence of the dampening effect rather than proof of its exclusive mechanism (Lauren Cohen and Andrea Frazzini, 2008).

4.8 Chapter Summary

This chapter has documented four interlocking findings. First, the full multimodal model M3 achieves AUC = 0.5655, delivering a +0.025 lift over the financial-only baseline — H1 confirmed. Second, and most critically, the M2 reversal (AUC −0.012 when undifferentiated FinBERT text is added) proves that naive NLP application to financial documents destroys predictive value, validating section-conditional parsing as a mathematical necessity, not merely a refinement — H2 empirically grounded before the OLS test. Third, bivariate OLS regression confirms the sign asymmetry ($\beta_{MDA} = +0.0044$, $\beta_{RF} = -0.0080$) on 1,140 deals — H2 formally supported. Fourth, Levene's test ($F = 7.07$, $p = 0.008$) and the significant negative correlation ($r = -0.070$, $p = 0.0002$) confirm that betweenness centrality acts as a

structural variance dampener on deal outcomes — H3 supported. The SHAP Global Manifold in the terminal converts the headline AUC gain into an economically interpretable finding: graph and text embeddings contribute non-trivially to predictions, and their value is structural and non-linear — exactly the properties that explain why prior finance-only models hit an accuracy ceiling that this architecture exits.

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Appendix A - Code

Use code blocks for snippets. Do not dump 1000s of lines here; use appendices for that. [cite: 20, 21]

```
static public void main(String[] args) {
  try {
    UIManager.setLookAndFeel(UIManager.getSystemLookAndFeelClassName());
  } catch(Exception e) {
    e.printStackTrace();
  }
  new WelcomeApp();
}
```

Listing 1: Java Example

Appendix B - Evaluation Include an evaluation of both product and process. Be objective. Reflect on what worked and what did not.

Appendix C - Conclusion Reflect on key findings, limitations, and future opportunities.

Appendix D - Initial Project Proposal

Appendix E - Final Project Proposal

Appendix F - Application for Approval of Research Activities

Appendix G - Client Consent Form